EE 324 - Controls Lab 2026

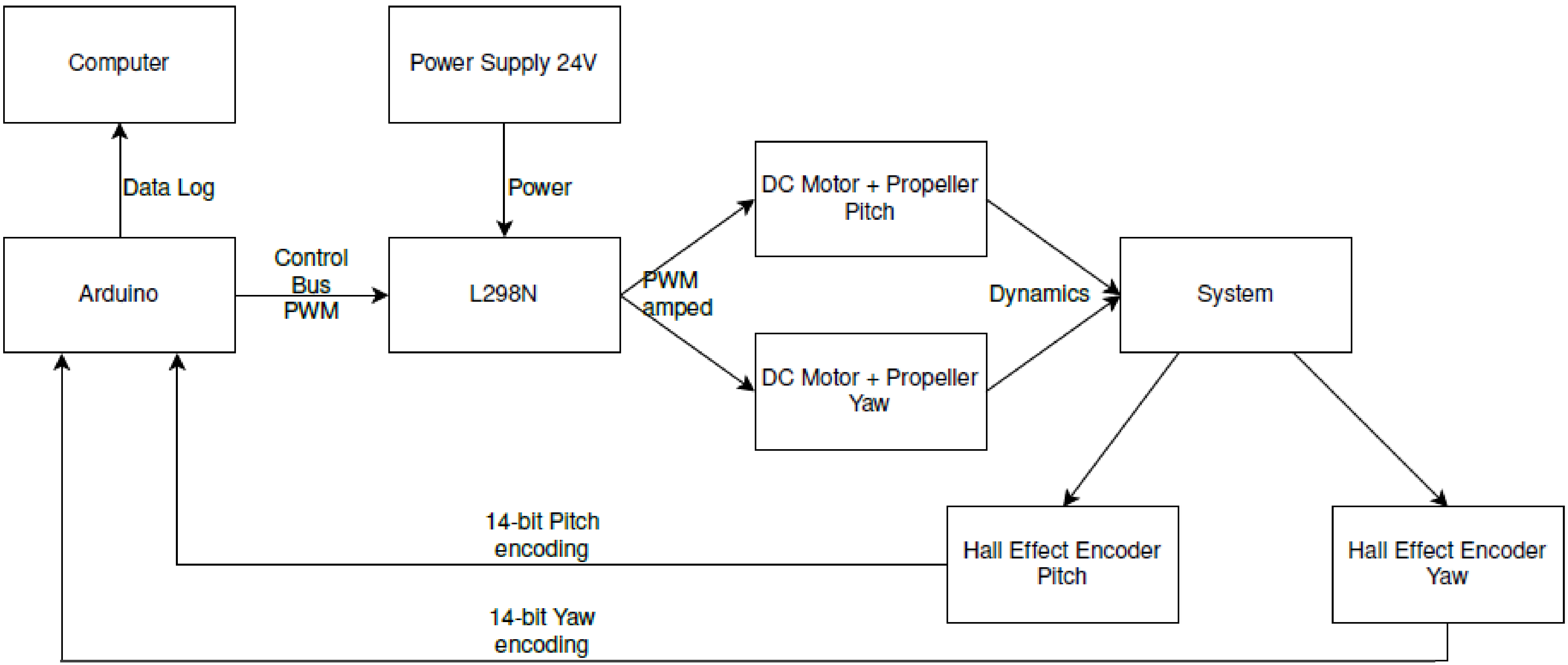
Lab Experiments

- Dual Axis Tilt Rotor Balancing Setup
- Line Follower
- Inverted Pendulum
- Noise Cancelling Headphone

Dual Axis Tilt Rotor Balancing Setup

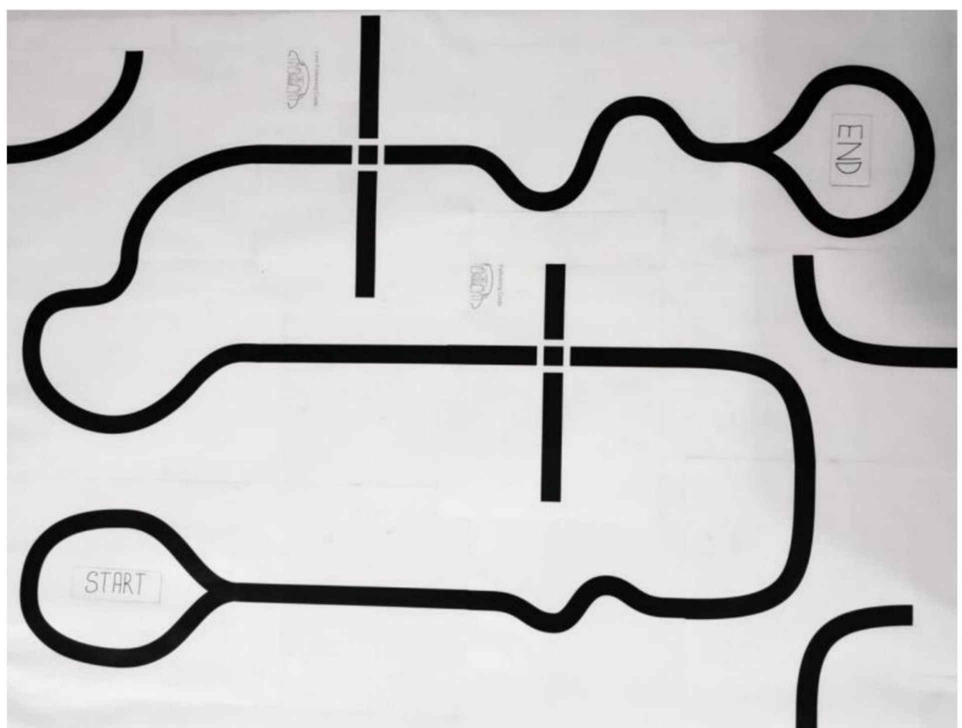
- Design and implement 2 PID feedback controller using Arduino Mega to control the pitch and Yaw position, simultaneously.

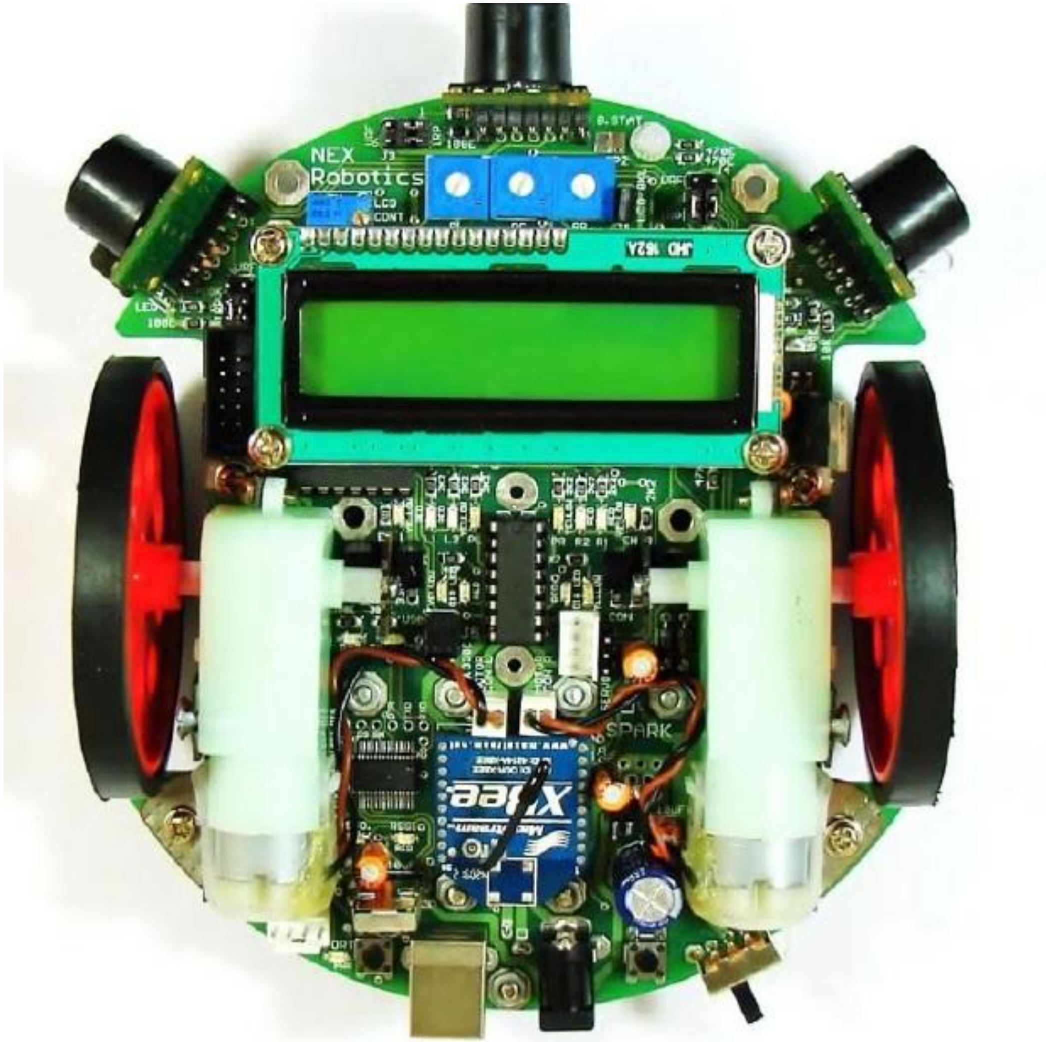
- Pitch:-
  - Peak overshoot: ≤ 5%
  - Peak time: ≤ 3.5 s
- Yaw:-
  - Peak overshoot: ≤ 5%
  - Peak time: ≤ 2.5 s


Computer
Power Supply 24V
Data Log
Power
DC Motor + Propeller
Pitch
Arduino
Control
Bus
PWM
L298N
PWM
amped
Dynamics
System
DC Motor + Propeller
Yaw
14-bit Pitch
encoding
Hall Effect Encoder
Pitch
Hall Effect Encoder
Yaw
14-bit Yaw
encoding

Line Follower

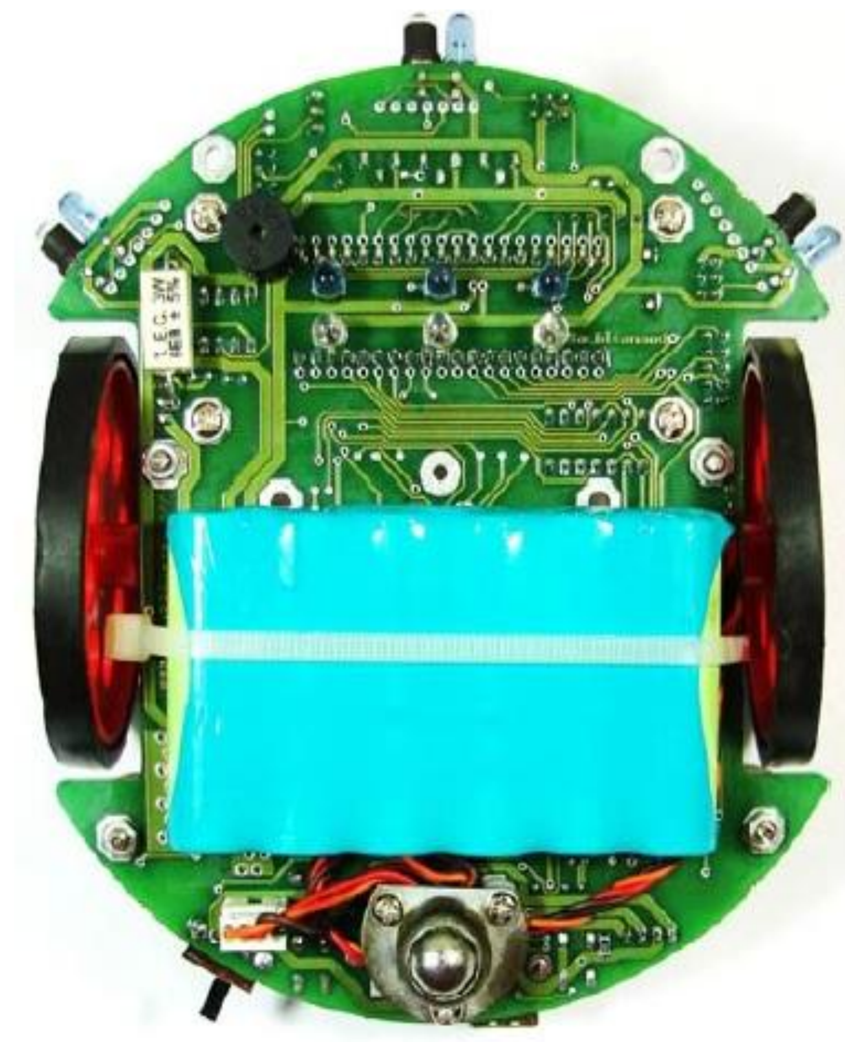
- Design and implement a PID controller for the Spark V robot to make it follow continuous lines (papers with printed sample lines will be provided) on the ground using the sensors provided on the robot for this purpose.





TOP VIEW

BOTTOM VIEW



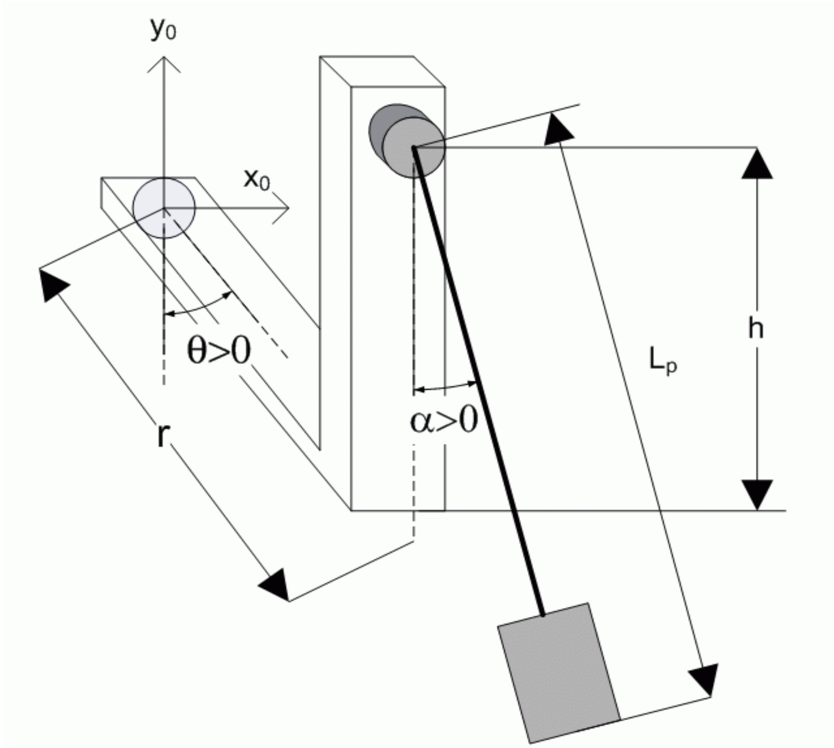
Inverted Pendulum

Design an LQR controller, i.e. tune the Q weighing matrix, such that the closed-loop response meets the following specifications:

(1) Arm Regulation: $|\theta(t)| < 30°$

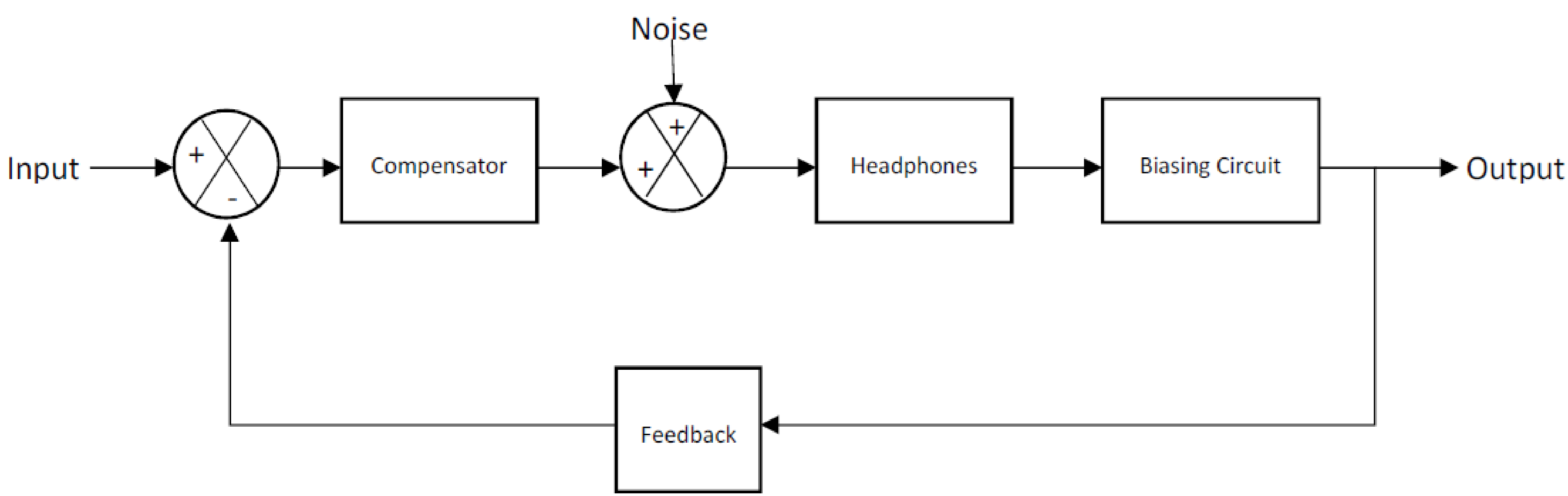
(2) Pendulum Regulation: $|\alpha(t)| < 3°$

(3) Control input limit: $V_m < 12$ V



Noise Cancelling Headphone

- Design and implement active noise cancellation in headphone using feedback system.
- Design to achieve a noise cancellation of 20 dB at 100 Hz while simultaneously maintaining a good stability margin.



Instructions

- Grades – Based on Experiments, Attendance, and End Sem Written Exam.
- Official and Extra Lab Sessions – Days and Timing.
- Demo & Viva – 2 TAs are required (Only on Official Session).
- Report Submission (1 for entire group) the week after viva and its format (next slide).
- Don’t take any lab equipment outside the lab.

Report Submission

- Upload to Moodle, zipped as follows: